

Helix Design Studio:

A Web Platform for Environment-Conditioned

Design of α -Helical Peptides

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Abstract

Rational design of α -helical peptides requires evaluating sequence compatibility within the physicochemical environment in which the helix is expected to function. This is particularly important for membrane-active and interfacial peptides, whose activity depends not only on intrinsic helical propensity but also on solvent polarity, electrostatics, membrane charge, and residue exposure. Here, we present *Helix Design Studio*, an open-source web platform for physics-based, environment-conditioned design and evaluation of α -helical peptides. The platform implements a parametric Hamiltonian that decomposes sequence–environment compatibility into physically interpretable contributions from hydrophobic partitioning, membrane-charge coupling, intrinsic and cooperative helical propensities, sequence-dependent pair interactions, and electrostatic interactions with environment-dependent screening. This unified model is used consistently for sequence scoring and optimization across homogeneous polar and apolar media, as well as anionic, neutral, and cationic interfaces. To enable quantitative assessment across sequences and environments, predicted energies are benchmarked against random-sequence ensembles matched for peptide length, yielding standardized *Z*-scores and percentile ranks. *Helix Design Studio* provides seven complementary design workflows spanning single-sequence evaluation, penetration-depth scanning, environment-specific optimization, diverse sequence-family generation, dual-environment comparison, one-vs-rest specificity design, and cross-environment compromise design. The platform also enables generation and export of idealized α -helical structures in PDB format for downstream visualization, docking, and molecular simulation. By integrating an interpretable physical objective with an accessible computational interface, *Helix Design Studio* provides a unified framework for translating environment-conditioned Hamiltonian models into practical peptide design workflows while retaining direct access to the physical contributions underlying each prediction and design decision.

1 Introduction and Motivation

The design of bioactive α -helical peptides, including antimicrobial peptides (AMPs), membrane-active agents, cell-penetrating peptides, and helix-bundle proteins, remains a central challenge in peptide engineering [1–3]. A major difficulty arises from the strong dependence of peptide behaviour on the physicochemical environment in which the helix is formed or operates [4]. In particular, membrane-active and interface-binding peptides must simultaneously accommodate hydrophobic partitioning, electrostatic interactions, residue exposure, and helix stability [5, 6]. Computational approaches that explicitly account for these environment-dependent effects can help prioritize sequences for experimental testing and reduce the search space associated with iterative sequence design, provided that their predictions are physically interpretable and sufficiently robust across relevant environments [7–9].

Recent advances in protein design have demonstrated the remarkable potential of data-driven methods for sequence generation and optimisation. For example, deep-learning approaches such as ProteinMPNN have achieved outstanding performance for the design of proteins with defined structural scaffolds [10]. However, most of these models have been developed and trained primarily on datasets dominated by water-soluble proteins. Their application to membrane-active and interface-binding helices therefore introduces additional challenges associated with the distinct physicochemical properties of non-aqueous and heterogeneous environments [8, 11]. Recent environment-conditioned and specialized generative approaches have begun to address this limitation by incorporating environmental information into the design process [3]. Nevertheless, their performance may remain strongly dependent on the structural representation and precision of the input scaffold [11]. More broadly, data-driven models primarily capture statistical relationships learned from their training distributions and do not necessarily provide a direct decomposition of the physical interactions [12] responsible for sequence–environment compatibility.

Mechanistic models provide a complementary perspective. Rather than learning sequence preferences exclusively from empirical distributions, physics-based formulations can encode the contributions of specific physicochemical interactions directly into the design objective. This provides an explicit connection between sequence identity, environmental conditions, and the resulting energetic preference [6]. Such an approach is particularly relevant for α -helical peptides, where the design objective is often not simply to maximize intrinsic helicity, but to identify sequences that are preferentially stabilized in a target environment relative to competing ones.

Helix Design Studio (Fig.1) implements this environment-conditioned perspective as an accessible computational platform. Translating a physics-based parametric Hamiltonian into a practical sequence-design framework, the tool does not predict a structure *de novo*. Instead, it evaluates the compatibility of a sequence with a predefined α -helical scaffold under a user-selected physicochemical environment. This formulation addresses a central design question:

How compatible is a given sequence with an α -helical state in a specific physicochemical environment?

This question forms the conceptual basis of the platform. To answer it, the studio integrates complementary workflows operating on the same underlying physical objective. Users

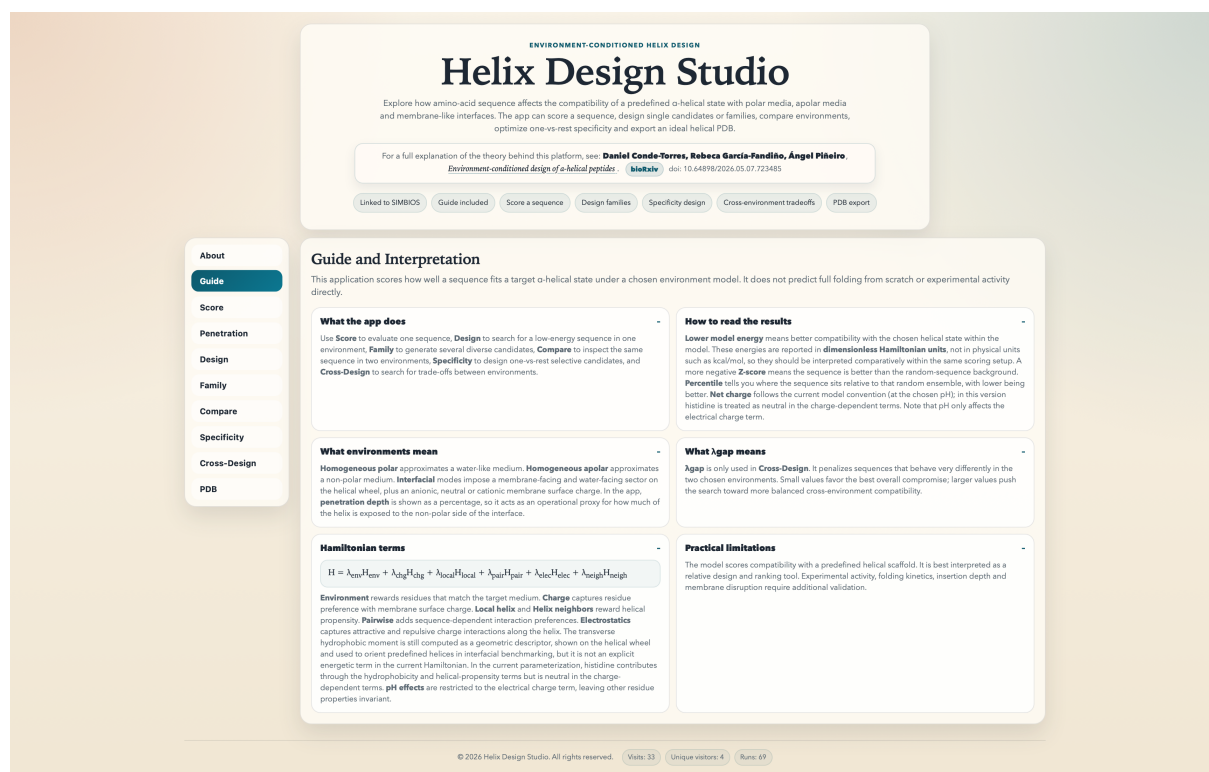


Figure 1: *Helix Design Studio* interface showing the navigation sidebar and main computational workspace.

can score existing references, scan for optimal membrane penetration depths, optimize sequences for a single environment, or generate diverse families of low-energy candidates. For more complex campaigns, the platform enables direct dual-environment comparisons, cross-environment compromise searches, and strict one-vs-rest specificity design. All results can be benchmarked against random-sequence ensembles via Z -scores and percentile ranks, and visualized through an integrated 3D idealized PDB export.

Importantly, the discrete formulation of the Hamiltonian provides a direct bridge to sequence optimization methods spanning classical and quantum-computing approaches. Because the physical objective is defined explicitly over amino acid identities, it can be explored using exact methods, classical heuristics, or hybrid quantum–classical strategies without requiring modification of the underlying physical model, as detailed in our companion study [6].

Developed by the SIMBIOS group¹ at the University of Santiago de Compostela, *Helix Design Studio* is conceived as a freely accessible resource for the scientific community. By unifying a physically interpretable Hamiltonian, quantitative benchmarking, and structural visualization within a single web-based interface without requiring local installation, the platform serves as a practical bridge between mechanistic peptide modeling and experimental sequence design.

¹<https://simbios.usc.es>

2 The Physical Hamiltonian

The computational core of *Helix Design Studio* is an environment-conditioned parametric Hamiltonian that quantifies the compatibility between an amino acid sequence and a pre-defined α -helical scaffold under a specified physicochemical environment. The formulation separates sequence–environment compatibility into six physically motivated contributions [6]:

$$\begin{aligned} \mathcal{H}(\mathbf{s}; \mathcal{E}) = & \lambda_{\text{pol}} H_{\text{pol}}(\mathbf{s}; \mathcal{E}) + \lambda_{\text{chg}} H_{\text{chg}}(\mathbf{s}; \mathcal{E}) + \lambda_{\text{local}} H_{\text{local}}(\mathbf{s}) \\ & + \lambda_{\text{pair}} H_{\text{pair}}(\mathbf{s}) + \lambda_{\text{elec}} H_{\text{elec}}(\mathbf{s}; \mathcal{E}) + \lambda_{\text{neigh}} H_{\text{neigh}}(\mathbf{s}), \end{aligned} \quad (1)$$

where $\mathbf{s}$ denotes the amino acid sequence and $\mathcal{E}$ defines the physicochemical environment. The coefficients λ_k are fixed relative weights, while each H_k represents a distinct physical contribution to the compatibility between the sequence and the prescribed helical state. The resulting Hamiltonian provides a common objective function for sequence evaluation, comparison, and optimisation throughout the platform.

Table 1: Components of the environment-conditioned Hamiltonian and their physical interpretation.

Term	Physical interpretation
H_{pol}	Hydrophobic partitioning, favouring apolar residues in membrane-like regions and polar residues in solvent-exposed regions, based on the Fauchère-Pliska scale [13].
H_{chg}	Coupling between residue charge and the local interfacial environment, favouring electrostatic complementarity between solvent-exposed residues and charged membrane surfaces.
H_{local}	Intrinsic per-residue propensity to adopt an α -helical conformation [14].
H_{neigh}	Cooperative sequence-dependent contribution involving neighbouring residues at $(i \pm 1)$, $(i \pm 3)$, and $(i \pm 4)$ positions along the sequence [15].
H_{pair}	Sequence-dependent pairwise interactions between side chains, parameterized via the Miyazawa-Jernigan contact matrix [16].
H_{elec}	Coulombic charge–charge interactions along the helical scaffold, modulated by environment-dependent dielectric screening [17].

The explicit decomposition of Eq. 1 is central to the interpretability of the platform. Unlike a purely statistical sequence model, the contribution of each physical component can be examined independently, while the physicochemical environment enters directly into the objective rather than being treated as an external annotation. A detailed mathematical derivation of the interaction potentials and the parameterization scheme is provided in the companion theoretical study [6].

2.1 Energy Normalization and Scoring Metrics

The raw Hamiltonian value provides the primary measure of sequence compatibility with the prescribed helical state. For a given scoring configuration, more negative values correspond to more favourable sequence–environment compatibility. Because the Hamiltonian is dimensionless and its absolute value depends on the selected scoring configuration, raw energies are interpreted comparatively rather than as absolute thermodynamic free energies.

To provide a standardized reference for sequence evaluation, the platform compares each sequence with an ensemble of randomly generated sequences of the same length. The corresponding standardized score is defined as

$$Z = \frac{\mathcal{H}_{\text{seq}} - \mu_{\text{rand}}}{\sigma_{\text{rand}}}, \quad (2)$$

where $\mathcal{H}_{\text{seq}}$ is the Hamiltonian value of the evaluated sequence, while μ_{rand} and σ_{rand} are the mean and standard deviation of the Hamiltonian distribution obtained from the random-sequence ensemble.

Negative Z -scores indicate that a sequence has a more favourable Hamiltonian than the corresponding random background, with increasingly negative values indicating stronger relative compatibility. The platform also reports the percentile rank of the sequence within the random distribution, providing an additional intuitive measure of its position relative to the reference ensemble.

The use of a length-matched random background is important because it converts the raw Hamiltonian into a relative statistical measure. Sequence quality can therefore be assessed in the context of the sequence space accessible under the same scoring conditions, rather than by interpreting the dimensionless Hamiltonian as an absolute physical quantity.

2.2 Hydrophobic Moment

The transverse hydrophobic moment, μ_H , is calculated according to the Eisenberg formalism [18] and is used as a geometric descriptor of the hydrophobic pattern projected onto the helical axis. Within *Helix Design Studio*, μ_H is displayed in the helical-wheel representation and provides a convenient description of sequence amphipathicity.

Importantly, μ_H is not included as an explicit energetic contribution to Eq. 1. It is therefore used for visualization, interpretation, and helix orientation in selected interfacial analyses, rather than as a direct optimisation term. Amphipathic organization observed in optimised sequences should consequently be interpreted as an emergent property of the combined Hamiltonian terms rather than as the result of explicitly optimising μ_H .

3 Physicochemical Environments

The physicochemical environment $\mathcal{E}$ is an explicit variable of the Hamiltonian and determines the environmental parameters and weights used to evaluate sequence–helix compatibility. *Helix Design Studio* provides five predefined environments spanning homogeneous polar and apolar media and charged or neutral membrane interfaces (Table 2).

Table 2: Predefined physicochemical environments available in *Helix Design Studio*.

Environment	Physical interpretation
Homogeneous polar	Water-like polar medium.
Homogeneous apolar	Non-polar, chloroform-like medium.
Interfacial anionic	Membrane-like interface with a negatively charged surface.
Interfacial neutral	Membrane-like interface with an electrically neutral surface.
Interfacial cationic	Membrane-like interface with a positively charged surface.

For interfacial environments, the penetration depth d is an operational model parameter controlling the position of the predefined helix relative to the polar–apolar boundary. It is expressed as a percentage ranging from 0% to 100%, where 0% corresponds approximately to a water-like configuration and 100% to a fully apolar configuration. This parameter should therefore be interpreted as a geometric control variable within the model rather than as a direct experimental measurement of membrane insertion depth.

4 Application Overview and Interface

Helix Design Studio provides a web-based interface that exposes the environment-conditioned Hamiltonian and its associated sequence-design workflows through a unified computational workspace. The interface is organised around a persistent navigation sidebar and a set of dedicated functional panels, allowing users to move seamlessly from sequence evaluation to environmental analysis, sequence optimisation, and structural export without changing the underlying physical model.

4.1 Navigation and Informational Panels

The application opens with a dedicated landing section introducing the scientific scope of the platform and its principal capabilities, providing users with immediate context and direct access to the associated scientific publication.

To support the interpretability of the model, a dedicated *Guide* panel provides contextual information and mathematical background to assist users in understanding the reported physical quantities. Institutional context is similarly provided in the *About* panel.

The core of the application, however, lies in its operational modules. Navigation is provided through a persistent left sidebar (Table 3), which separates the platform into sequence evaluation, environmental analysis, sequence optimisation, comparative design, and structural export.

Table 3: Navigation structure of *Helix Design Studio*.

Module	Function
<i>About</i>	Institutional and scientific context.
<i>Guide</i>	Scientific interpretation of the model and outputs.
<i>Score</i>	Evaluation of a single sequence in a selected environment.
<i>Penetration</i>	Analysis of sequence compatibility as a function of interfacial penetration depth.
<i>Design</i>	Single-environment sequence optimisation.
<i>Family</i>	Generation of diverse low-energy peptide sequences.
<i>Compare</i>	Quantitative comparison of a sequence across two environments.
<i>Specificity</i>	One-vs-rest optimisation for environment-specific sequence design.
<i>Cross-Design</i>	Optimisation of sequence compatibility across multiple environments.
<i>PDB</i>	Generation and export of idealised α -helical structures.

5 Operational Modules

The computational workflows implemented in *Helix Design Studio* all operate on the environment-conditioned Hamiltonian defined in Eq. 1. The individual modules differ in how this common objective is used: sequences can be evaluated directly, analysed as a function of penetration depth, optimised for a single environment, compared across environments, designed for environmental specificity, or optimised across multiple environments. This section describes the principal operational workflows exposed through the application interface.

5.1 *Score* — Sequence Evaluation

The *Score* module provides direct evaluation of an input peptide sequence under a user-selected physicochemical environment. Sequences of up to 40 residues can be entered using the standard one-letter amino acid code, while the user can specify the environmental preset, pH, penetration depth for interfacial environments, and the size of the random reference ensemble.

The sequence is evaluated using the Hamiltonian defined in Eq. 1, and the resulting energy is reported together with its standardized Z -score and percentile relative to a random background of sequences with the same length. The random reference ensemble can contain between 100 and 2000 sequences, allowing the user to control the resolution of the statistical reference distribution.

For illustration, the module can be used to evaluate experimentally characterised reference peptides such as Magainin 2 under interfacial anionic conditions. The resulting score provides a quantitative measure of the compatibility of the sequence with the predefined α -helical scaffold under the selected environment (Fig. 2).

5.1.1 Built-in Reference Peptide Library

To facilitate benchmarking and rapid exploration, the application provides a curated library of reference sequences spanning soluble helical peptides, amphipathic membrane-active peptides, and transmembrane model systems. These sequences provide predefined

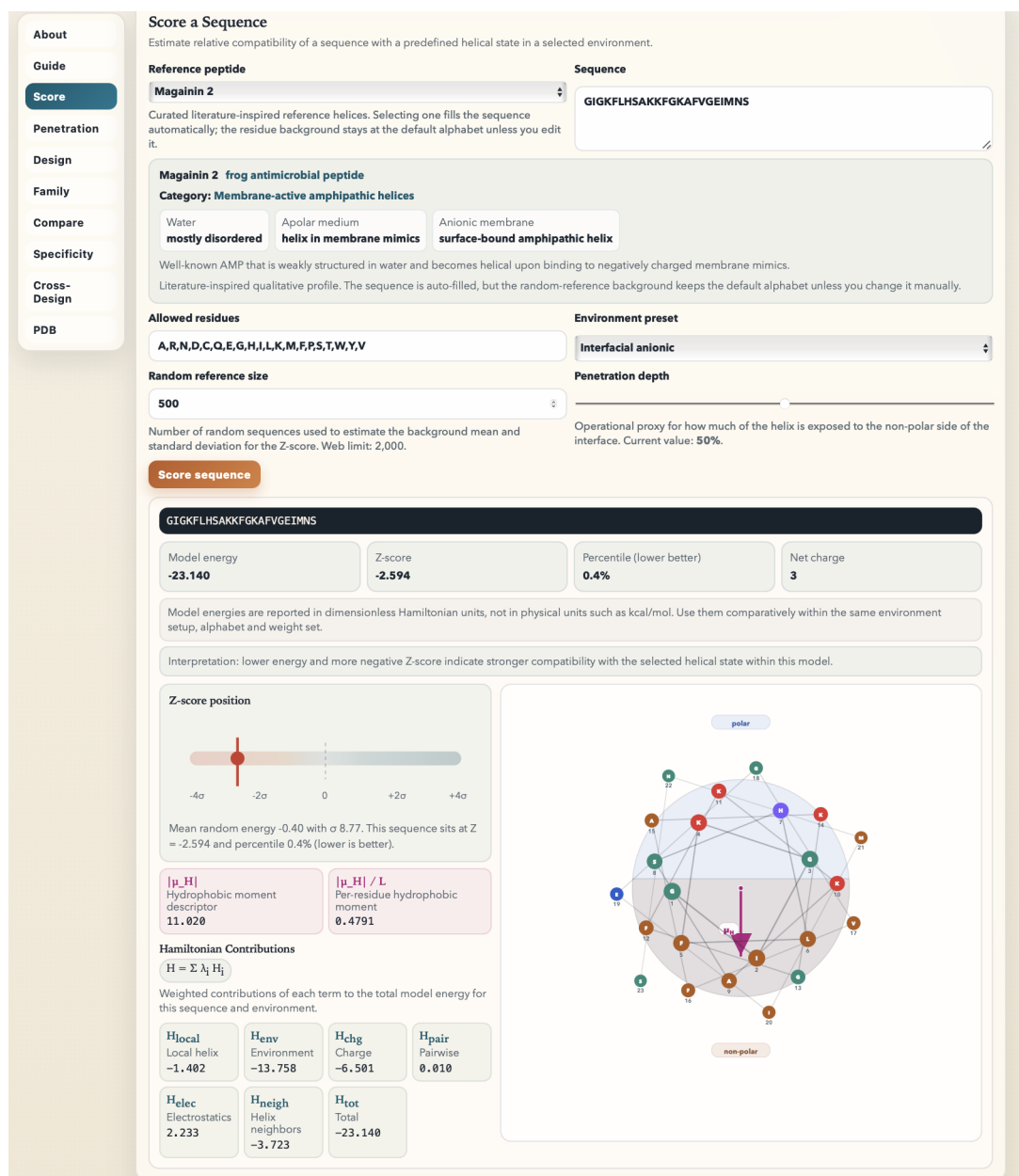


Figure 2: *Score* module showing the controls available for environment-conditioned sequence evaluation.

examples that can be evaluated without manual sequence entry [6]:

- **Water-soluble and helix-forming controls:** (EAAAK)₃ [19], (AAKAA)₃ [20].
- **Amphipathic membrane-active helices:** LL-37 [21], Melittin [22], Magainin 2 [23], Cecropin A [24], PGLa [25].
- **Transmembrane helix model peptides:** WALP23 [26], KALP17 [27].

The diversity of the reference library allows the same Hamiltonian to be examined across substantially different sequence classes and physicochemical contexts. These reference sequences are intended primarily as benchmark and demonstration cases rather than as a training set for the model.

5.1.2 Score Module Outputs

For each evaluated sequence, the *Score* module reports three quantitative measures derived from the Hamiltonian and its random reference distribution:

- The raw Hamiltonian energy, together with the corresponding Z -score and percentile;
- A helical-wheel representation displaying the transverse hydrophobic moment μ_H .
- A distribution plot showing the position of the evaluated sequence relative to the random-sequence energy background.

Together, these outputs provide complementary views of sequence compatibility. The Hamiltonian provides the underlying physical score, while the Z -score and percentile place that score in the context of the corresponding random sequence space. The helical-wheel representation provides an independent geometric description of the sequence and its hydrophobic organisation.

5.2 *Penetration* — Depth-of-Insertion Scan

The *Penetration* module evaluates the dependence of sequence–environment compatibility on the position of an α -helical scaffold relative to an interfacial polar–apolar boundary. For a selected interfacial environment, the platform automatically evaluates the sequence at penetration depths ranging from 0% to 100% in 5% increments. The penetration depth is treated as an operational geometric parameter of the model, as defined in Section 3.

For each depth, the sequence is evaluated using the same environment-conditioned Hamiltonian and the corresponding random-sequence reference distribution. The module identifies the depth associated with the most negative Z -score and reports the complete penetration profile, allowing the dependence of sequence compatibility on insertion depth to be examined rather than requiring the user to specify an optimal depth in advance.

As an illustrative example, a penetration scan of Magainin 2 under an interfacial anionic environment identifies a minimum at 30% penetration, with a corresponding Z -score of -3.673 (Fig. 3). This example demonstrates how the module can be used to identify the preferred position of a predefined helical scaffold within the simplified interfacial model.

5.3 *Design* — Single-Environment Sequence Optimisation

The *Design* module performs sequence optimisation for a single target physicochemical environment. The optimisation operates directly on the discrete amino acid sequence space using classical simulated annealing [28]. At each Monte Carlo step, a residue is selected for mutation and the resulting sequence is evaluated using the Hamiltonian defined in Eq. 1. The temperature is progressively reduced according to the selected cooling schedule, allowing the search to explore the energy landscape before increasingly favouring low-energy regions.

The principal configuration options available across the operational modules are summarised in Table 4.

Multiple independent restarts can be used to improve exploration of the sequence landscape and reduce dependence on a single initial condition. The final result is selected from the solutions obtained across the independent runs according to the optimisation objective.

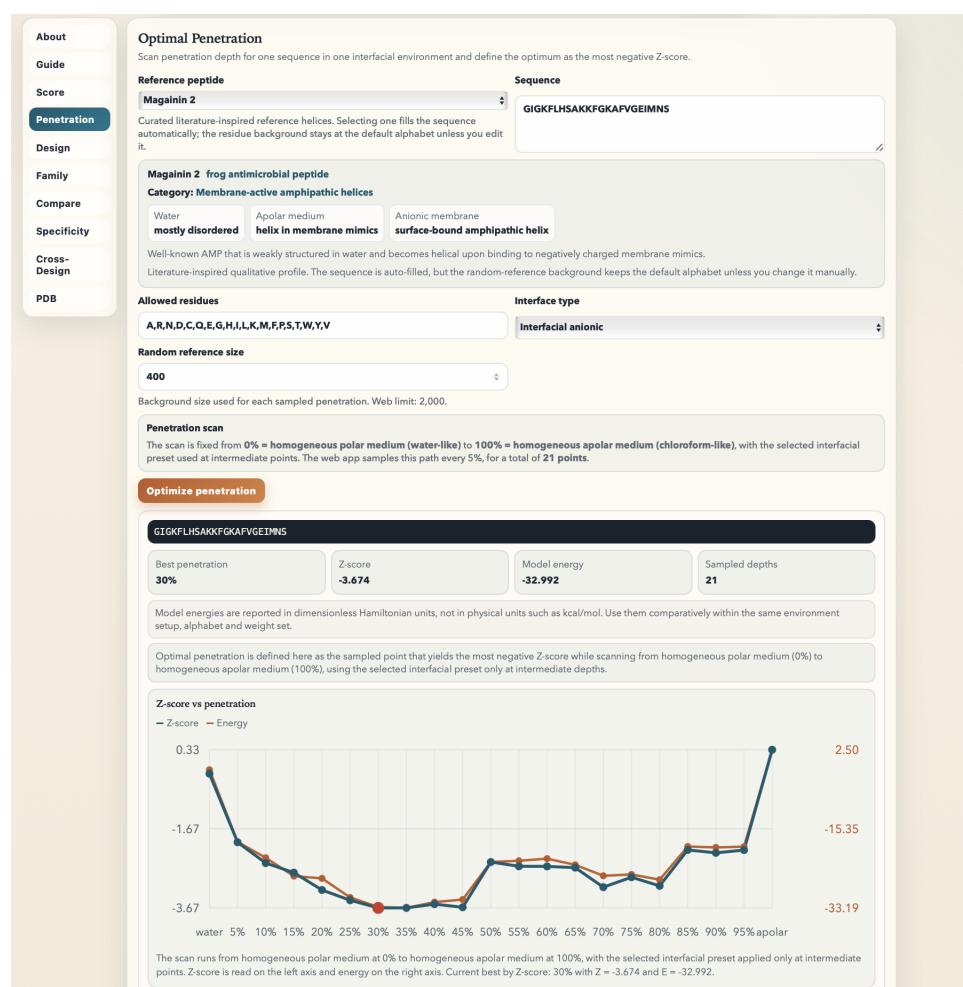


Figure 3: *Penetration* module showing the depth-dependent evaluation of sequence compatibility across an interfacial environment.

As an illustrative design example, a 20-residue sequence optimised for an interfacial anionic environment converges to a highly amphipathic sequence. The resulting design has a Z-score of -8.709 , corresponding to a very favourable position within the random-sequence reference distribution (Fig. 4). The Hamiltonian decomposition provides additional information about the physical contributions underlying the score, with favourable environmental and electrostatic terms contributing substantially to the final objective. The corresponding helical-wheel representation provides a complementary visualisation of the resulting hydrophobic organisation.

5.4 *Family* — Diverse Peptide Family Generation

The *Family* module extends single-sequence optimisation by generating multiple low-energy candidates while explicitly promoting sequence diversity. Rather than returning only the lowest-energy solution, the module performs independent optimisation runs to obtain a set of distinct candidate sequences for a common target environment.

The objective is therefore not to identify a single global optimum, but to provide a collection of energetically favourable sequences occupying different regions of sequence space. This can be useful when several candidates are required for downstream experimental screening

Table 4: Configuration options available across the operational modules of *Helix Design Studio* and their corresponding web-interface limits.

Control	Range/Limit	Description
Environment preset	Dropdown / Five presets	One of the five physicochemical environments defined in Table 2.
pH	0–14	Environmental parameter affecting the electrostatic charge calculation.
Penetration depth	0–100%	Position of the helix relative to the polar–apolar interface; available for interfacial environments.
Reference peptide	Dropdown	Pre-loaded library of experimentally characterised or commonly used reference sequences.
Sequence	Text, max 40 res.	Amino acid sequence specified using the one-letter code.
Random reference size	Variable limit	Number of random sequences used to construct the reference energy distribution.
Length	4–40	Number of residues in the designed sequence.
Allowed residues	User-defined	Comma-separated one-letter amino acid codes permitted during optimisation.
Family size	2–12	Number of distinct candidate sequences requested.
Optimisation steps	Variable limit	Maximum number of Monte Carlo mutation steps per optimisation run.
Restarts	Variable limit	Number of independent optimisation runs initiated from different starting sequences.

or when sequence diversity is desired in addition to energetic performance.

The number of requested family members, optimisation steps, and independent restarts per candidate can be configured through the options previously summarised in Table 4.

The module attempts to maintain sequence distinctness while retaining favourable Hamiltonian values. If the specified diversity requirements cannot be satisfied within the explored sequence space, the returned family may contain fewer candidates than requested.

As shown in Fig. 5, the module can generate a collection of distinct sequences with comparable favourable energy scores. The resulting family illustrates the ability of the platform to explore multiple low-energy regions of sequence space rather than focusing exclusively on a single optimisation trajectory. This provides a practical strategy for generating alternative candidates for subsequent analysis or experimental evaluation.

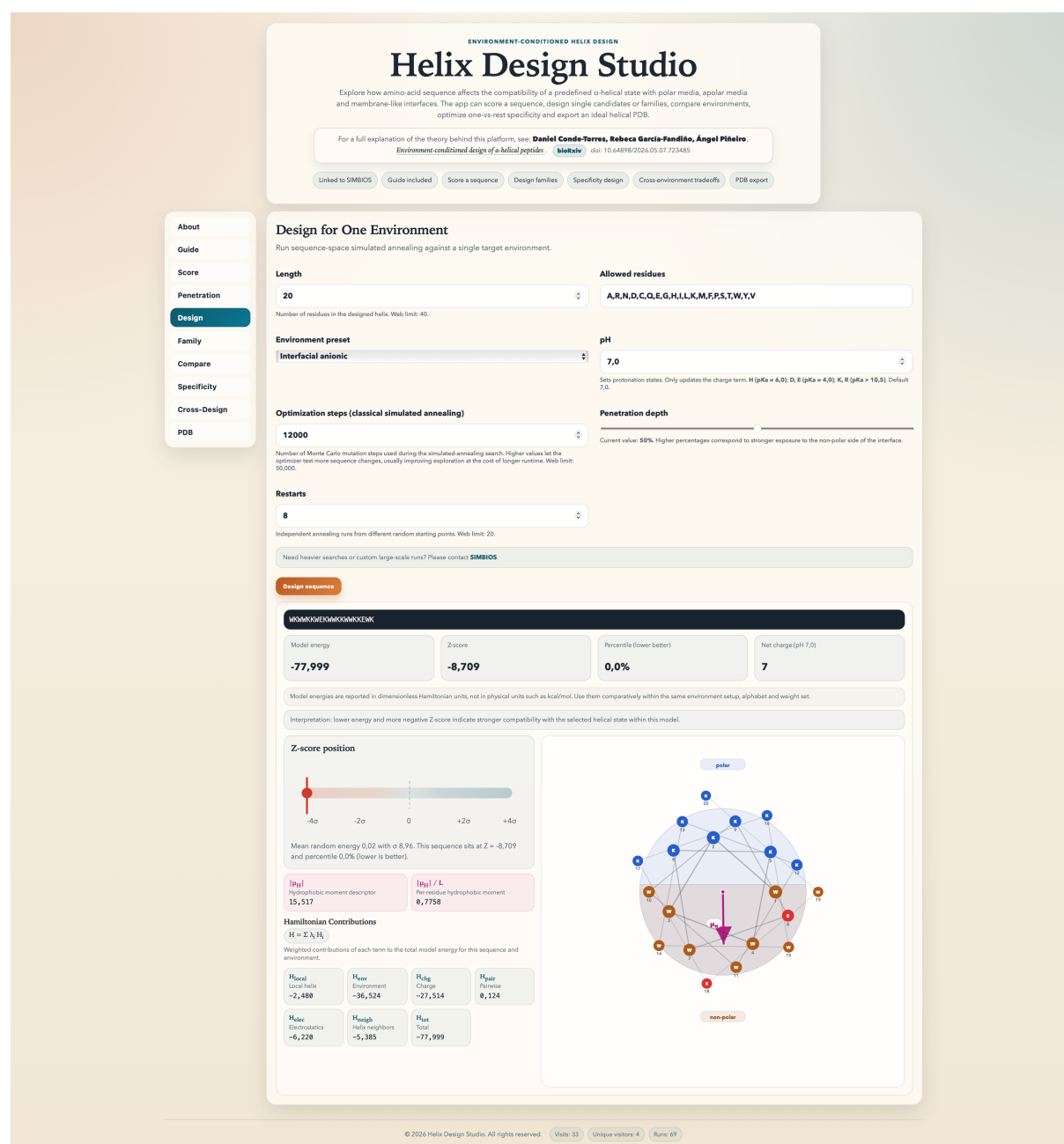


Figure 4: Example output from the *Design* module for a 20-residue sequence optimised for an interfacial anionic environment. The result includes the optimised sequence, Hamiltonian metrics with random-sequence benchmarking, and a helical-wheel representation of the resulting sequence.

5.5 Compare — Dual-Environment Comparison

The *Compare* module evaluates the same amino acid sequence under two independently selected physicochemical environments. This configuration allows the environmental dependence of sequence compatibility to be assessed without changing the primary sequence between conditions. For interfacial environments, each condition can be assigned an independent penetration depth, enabling comparisons between different positions of the helix relative to the corresponding interfaces.

About
Guide
Score
Penetration
Design
Family
Compare
Cross-Design
PDB

Design a Family

Generate several unique low-energy candidates for the same environment instead of a single optimum.

Length
 20
 Number of residues in each designed helix. Web limit: 40.

Allowed residues
 A,P,D,E,Q,L,M,F,K,S,T,W,Y,V,H,G

Environment preset
 Interfacial anionic

Family size
 6
 Requested number of diverse low-energy candidates. If the search cannot find that many unique solutions, the returned family may be smaller. Web limit: 12.

Optimization steps (classical simulated annealing)
 10000
 Number of Monte Carlo mutation steps used in each annealing run. More steps generally help each candidate settle into a better low-energy sequence, but they also make family generation slower. Web limit: 20,000.

Penetration depth
 Current value: 50%. Used only for interfacial families and interpreted as non-polar exposure percentage.

Restarts per candidate
 6
 More restarts improve robustness but increase runtime. Web limit: 20.

Need larger families or heavier searches? Please contact [SIMBIOS](#).

Generate family

Download JSON

Requested: 6 Returned: 6 Environment: interfacial_neg Alphabet size: 16

Generating candidates 6/6 accepted after 6 attempts
 Latest attempt: WKWEKKWKKWKKWEKKWFK

Generated 6 unique candidates

Candidate 1 Energy -83.748 | Z -9.207
 WKWKKWKKWEKKWKKWEK

Candidate 2 Energy -83.729 | Z -9.205
 WKWEKKWKKWKKWEKKWKK

Figure 5: Example output from the *Family* module showing multiple diverse, low-energy candidate sequences optimised for an interfacial anionic environment.

The Hamiltonian energy, standardized Z -score, percentile, and helical-wheel representation are displayed side-by-side for the two environments. This parallel representation provides a direct means of quantifying the energetic difference associated with changing the physicochemical context while holding sequence identity constant.

As an illustrative example, the custom sequence WFEKWEKEWKEFWKEWKDWF can be evaluated under homogeneous polar and homogeneous apolar conditions (Fig. 6). The resulting comparison shows a more favourable Hamiltonian value under the homogeneous apolar condition. This is expected, since the sequence has a high propensity to form α -helices, and the apolar environment is more favourable for the formation of α -helices. The example illustrates how the *Compare* module can be used to distinguish sequence compatibility arising from environmental context rather than from differences in sequence composition.

5.6 Specificity — One-vs-Rest Design

The *Specificity* module extends single-environment sequence optimisation by introducing an explicit objective for environmental discrimination. The design problem is formulated as a one-vs-rest optimisation in which the sequence is encouraged to achieve favourable com-

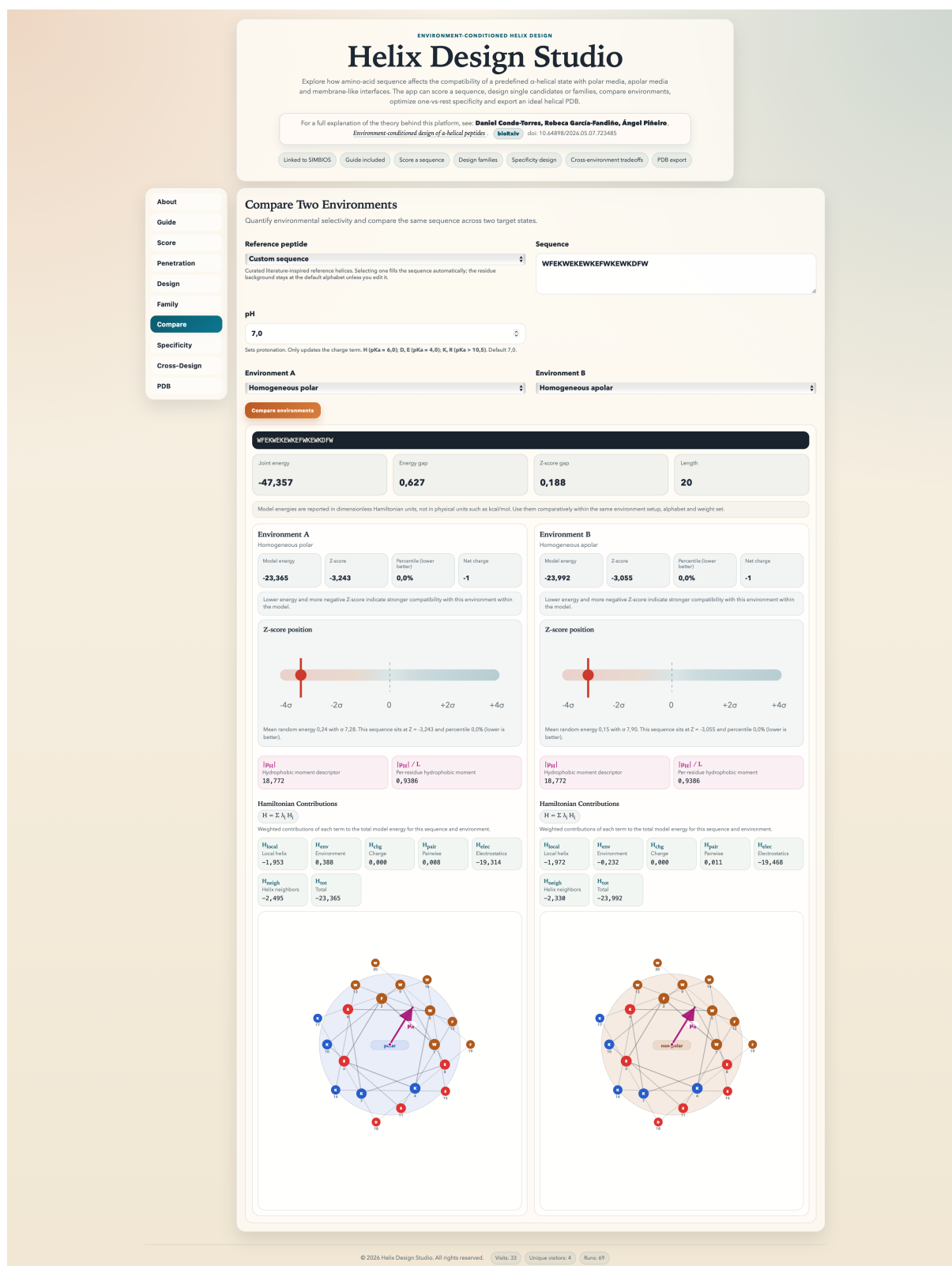


Figure 6: *Compare* module showing the evaluation of the custom sequence WFEKWEKEFWKEWKDWF under two physicochemical environments. The side-by-side representation reports the corresponding Hamiltonian metrics and helical-wheel visualisations.

patibility in a designated target environment while simultaneously reducing compatibility with one or more selected off-target environments.

The corresponding objective function is

$$\mathcal{L}_\lambda = \lambda H_{\text{target}} + (1 - \lambda) \left(H_{\text{target}} - \min_k H_{\text{off-target},k} \right), \quad (3)$$

where H_{target} denotes the Hamiltonian evaluated in the target environment, $H_{\text{off-target},k}$ denotes the Hamiltonian obtained in off-target environment k , and $\lambda \in [0, 1]$ controls the relative importance assigned to target compatibility and one-vs-rest discrimination.

The formulation therefore introduces an explicit trade-off between absolute performance in the target environment and the energetic separation from the most favourable off-target condition. In the limiting cases,

- $\lambda \rightarrow 1$ places increasing emphasis on the absolute Hamiltonian of the target environment.
- $\lambda \rightarrow 0$ places increasing emphasis on the one-vs-rest discrimination objective.

The target environment is automatically excluded from the off-target set, and up to four off-target environments can be selected. During optimisation, candidate sequences are therefore evaluated simultaneously against the target and the selected off-target conditions.

The resulting candidates are ranked according to the specificity objective, while the individual environmental scores are retained to allow the user to distinguish target compatibility from off-target rejection. An illustrative optimisation targeting an anionic interface while penalising compatibility with a homogeneous polar environment is shown in Fig. 7. The resulting candidates exhibit favourable target scores together with reduced compatibility in the selected off-target condition, demonstrating the intended behaviour of the one-vs-rest objective.

5.7 *Cross-Design* — Cross-Environment Compromise

The *Cross-Design* module addresses a complementary optimisation problem: instead of maximising discrimination between environments, it seeks sequences that maintain favourable compatibility across two environments simultaneously (Fig. 8). This formulation is intended for cases in which a sequence must operate across distinct physicochemical contexts rather than being selective for a single target condition.

Let H_A and H_B denote the Hamiltonian values of a candidate sequence in environments A and B , respectively. The cross-environment objective is

$$\mathcal{H}_{\text{total}} = H_A + H_B + \lambda_{\text{gap}} |H_A - H_B|, \quad (4)$$

where λ_{gap} controls the penalty associated with differences in energetic performance between the two environments.

The first two terms favour sequences with favourable combined energy across the two conditions, whereas the absolute-difference term penalises imbalanced solutions. Consequently:

Figure 7: Example output from the *Specificity* module for an anionic target environment and a homogeneous polar environment as the off-target environment. Candidate sequences are ranked according to the specificity objective and accompanied by their individual environmental scores.

- Small λ_{gap} values place greater emphasis on the combined Hamiltonian and therefore favour the best overall energetic compromise.
- Larger λ_{gap} values increasingly penalise differences between H_A and H_B , favouring more balanced performance across the two environments.

This formulation provides a direct counterpart to the *Specificity* module: whereas *Specificity* seeks energetic separation between a target and off-target environments, *Cross-Design* seeks compatibility without strong energetic imbalance between two selected environments.

5.8 PDB — Idealised Helix Structure Export

The *PDB* module converts an amino acid sequence into an idealised α -helical structure that can be exported in Protein Data Bank (PDB) format (Fig. 9). Sequences of up to 40 residues can be processed, providing a convenient structural representation for subsequent visualisation, molecular modelling, docking, or molecular dynamics workflows.

The structures are generated using Biopython [29] and PeptideBuilder [30]. The backbone is constructed using canonical α -helical dihedral angles [31]:

$$\phi = -57^\circ, \quad \psi = -47^\circ, \quad (5)$$

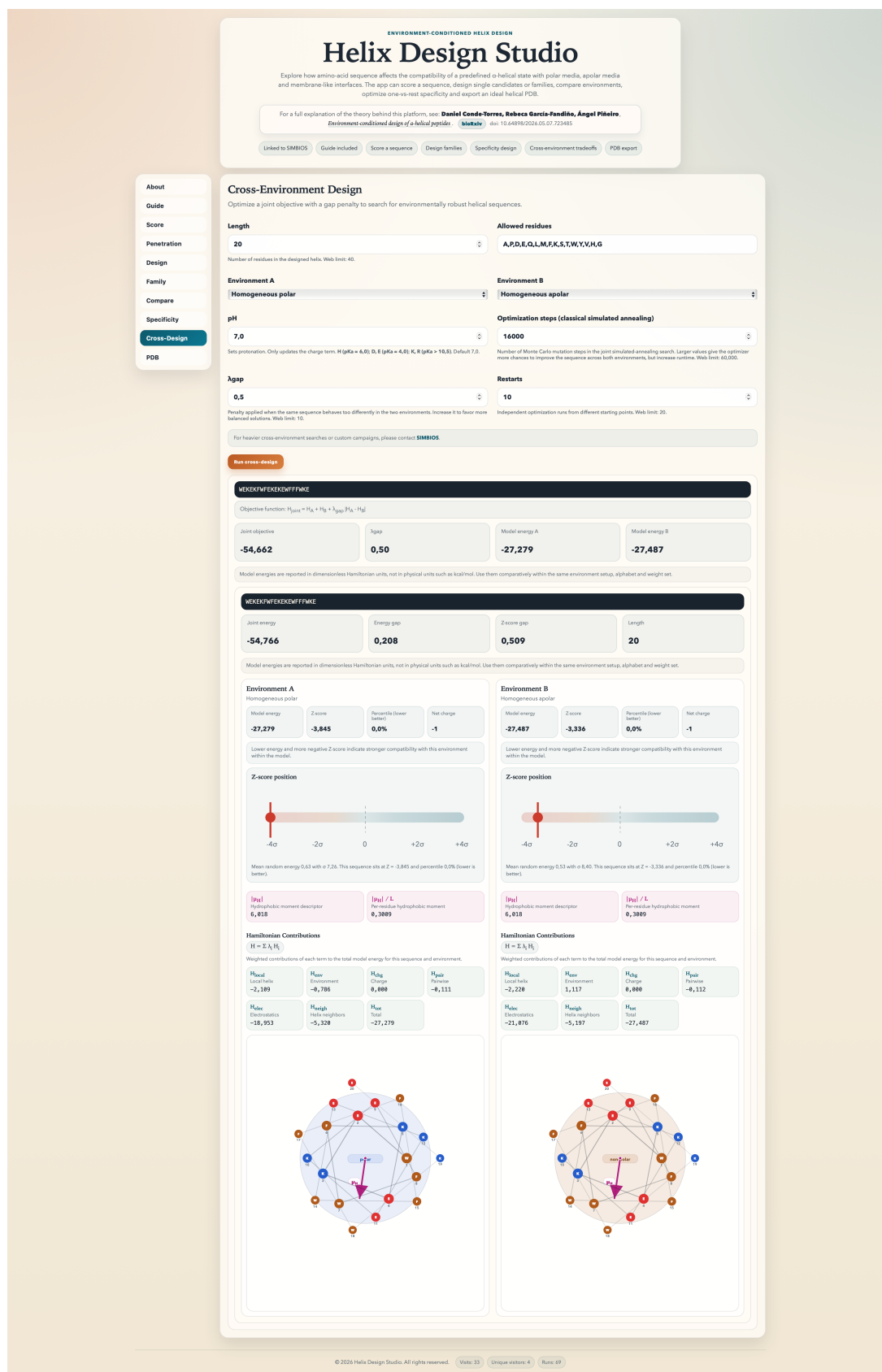


Figure 8: Cross-Design module showing the λ_{gap} parameter used to control the balance between overall compatibility and energetic similarity across two environments.

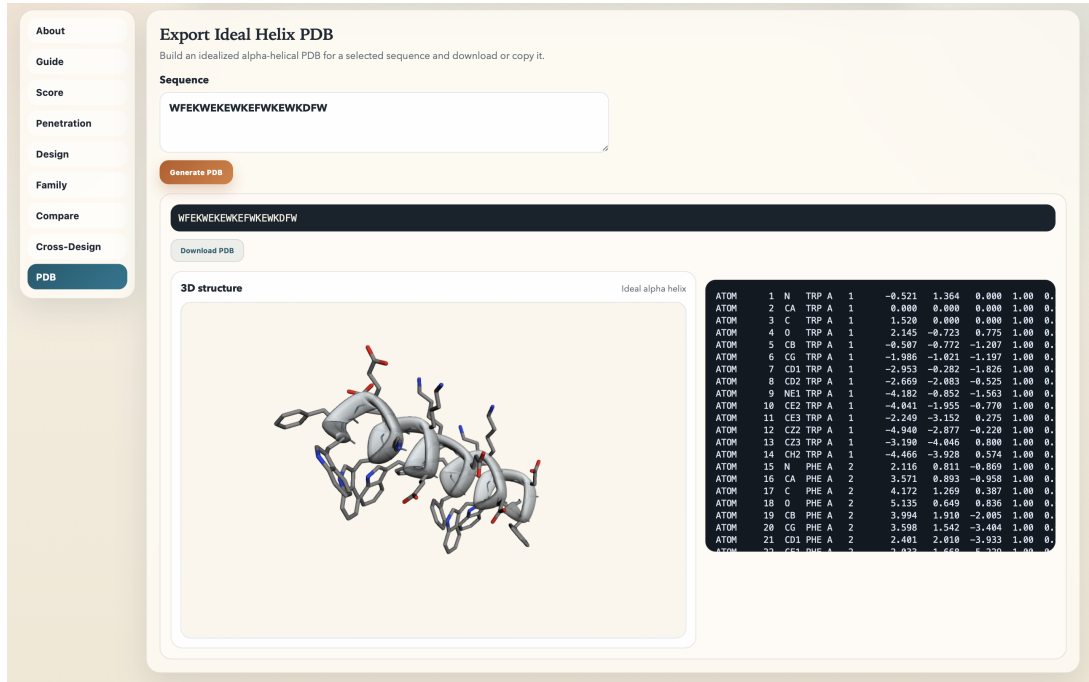


Figure 9: *PDB* module for generation, export, and interactive visualisation of idealised α -helical structures.

Together with idealised side-chain conformations. The resulting structure should be interpreted as an idealised starting geometry rather than as a predicted experimentally resolved structure or an energy-minimised conformation.

An integrated `3Dmol.js` viewer [32] provides immediate browser-based inspection of the generated structure. The visualisation includes:

- Cartoon representation of the helical backbone.
- Stick representation of side chains using standard Jmol element colouring.
- Automatic centring with interactive zoom and rotation.

The combination of PDB export and browser-based visualisation provides a direct connection between the sequence-design workflows and downstream structural modelling. The exported structures can therefore serve as standardised initial geometries for subsequent computational analyses, while retaining the distinction between the idealised starting scaffold generated by the platform and any structure subsequently obtained through higher-resolution modelling.

6 Interpretation Guide

The Guide panel provides contextual information to support the interpretation of the computational outputs generated by *Helix Design Studio*. Rather than introducing an independent analysis method, the Guide serves as an integrated reference for the physical model, the environmental parameters, and the scoring metrics described throughout this manuscript.

The available information is organised into expandable sections covering:

1. The purpose and operation of each computational module.
2. Interpretation of Hamiltonian energies, Z -scores, and percentile values.
3. Definitions of the physicochemical environments and the interfacial penetration-depth parameter.
4. The mathematical formulation and interpretation of the environment-conditioned Hamiltonian in Eq. 1.

Several interpretive rules are particularly important when analysing platform outputs:

- Hamiltonian values are expressed in dimensionless model units and should be interpreted comparatively within a common scoring configuration. They should not be interpreted as absolute thermodynamic free energies.
- The pH parameter affects the electrostatic charge contribution H_{elec} only. Other residue-specific physicochemical properties represented in the current model are treated as pH-independent.

These conventions are explicitly communicated through the Guide to reduce misinterpretation of the model outputs and to distinguish the quantities calculated by the Hamiltonian from experimentally measurable thermodynamic or structural properties.

7 Technical Architecture

7.1 Technology Stack

Helix Design Studio separates the user interface from the scientific computation through a lightweight client–server architecture. The frontend provides the interactive interface and visualisation layer, while the backend executes the sequence scoring, optimisation, and structure-generation routines. The principal software components are summarised in Table 5.

Table 5: Software stack of *Helix Design Studio*.

Layer	Technology
Backend server	FastAPI [33] and Uvicorn (Python)[34]
Frontend	HTML5, CSS3, and JavaScript
Scientific engine	Python [35] and NumPy [36]
PDB generation	Biopython [29] and PeptideBuilder [30]
3D visualisation	3Dmol.js [32]
Deployment	Docker [37] and Azure Container Apps [38]

The scientific engine is implemented in Python and provides the common computational layer used by the different application modules. This separation allows the same physical scoring functions to be accessed through multiple workflows while maintaining a consistent implementation of the underlying Hamiltonian.

7.2 REST API

Communication between the frontend and the computational backend is implemented through a RESTful API. Each major workflow is exposed through a dedicated endpoint, allowing the web interface to act as a client of the same underlying computational services. The principal endpoints are listed in Table 6.

Table 6: Principal REST API endpoints implemented by *Helix Design Studio*.

Endpoint	Function
POST /api/score	Sequence scoring in a selected environment.
POST /api/penetration	Automated penetration-depth scan.
POST /api/design	Single-environment sequence optimisation using simulated annealing.
POST /api/family	Generation of diverse low-energy sequence candidates.
POST /api/compare	Dual-environment sequence comparison.
POST /api/specificity	One-vs-rest specificity sequence design.
POST /api/cross	Cross-environment compromise optimisation.
POST /api/pdb	Idealised α -helical PDB generation.
GET /api/health	Service health check.
GET /api/usage-metrics	Retrieval of application usage statistics.
POST /api/visit	Anonymous application-visit registration.

The API-based architecture separates the scientific computation from the presentation layer and provides a consistent interface through which the different optimisation workflows can be accessed. This structure also allows the computational engine to be invoked independently of the particular frontend implementation.

7.3 Streaming Output

Several optimisation workflows can require substantially more computation than a single sequence evaluation. To avoid requiring the browser to wait for completion of the entire optimisation process, the /api/family and /api/specificity endpoints return intermediate results using Newline-Delimited JSON (NDJSON) over HTTP streaming.

Candidate solutions can therefore be transmitted to the client progressively as they are generated and rendered incrementally in the user interface. This approach provides immediate feedback during computationally longer workflows while preserving the same underlying server-side optimisation procedure.

7.4 Deployment and Data Handling

The application is containerised using Docker and deployed through Azure Container Apps. The current implementation does not require an external database for the scientific

workflows. Anonymous application usage events, including visit and run statistics, are recorded in a local `.jsonl` file and used to provide aggregate usage information within the application.

The usage-tracking layer is therefore separate from the scientific scoring and optimisation pipeline. It does not form part of the Hamiltonian calculation or affect the sequence-design objective.

8 Known Limitations

The capabilities of *Helix Design Studio* should be interpreted in the context of the assumptions and approximations underlying its physical model. Several limitations are therefore important when using the platform for sequence design or biological interpretation [6].

- The model evaluates sequence compatibility with a predefined α -helical scaffold. It does not perform *de novo* protein folding, backbone conformational sampling, or unrestricted structure prediction.
- Hamiltonian energies are expressed in dimensionless model units and are meaningful primarily for comparative analyses performed under the same scoring configuration.
- The interfacial penetration parameter is an operational model variable rather than a direct experimental measurement of membrane insertion depth.
- Predictions of experimental activity, membrane binding, folding kinetics, membrane disruption, permeability, toxicity, or biological function require independent experimental validation.
- The current treatment of pH affects only the electrostatic contribution. Other residue-specific physicochemical properties are assumed to remain unchanged with pH.
- The environmental presets represent simplified physicochemical conditions and should not be interpreted as complete molecular models of biological membranes or cellular environments.
- The optimisation procedures operate within finite sequence lengths, mutation spaces, numbers of steps, and restart limits imposed by the web application. These constraints limit the extent of sequence space that can be explored in a single run.
- The absence of an external database in the current deployment simplifies the computational architecture but also means that the platform does not maintain a persistent experimental-results database or an integrated mechanism for automatically incorporating new experimental measurements.

These limitations define the intended scope of the platform: it is a mechanistic sequence-design and hypothesis-generation framework rather than a substitute for experimental characterisation or high-resolution molecular simulation.

9 Conclusion

Helix Design Studio provides an environment-conditioned framework for the evaluation and design of α -helical peptide sequences. Its central component is a physically motivated Hamiltonian that decomposes sequence–environment compatibility into interpretable contributions, including environmental partitioning, electrostatic interactions, intrinsic and cooperative helical propensity, pairwise side-chain interactions, and charge–charge interactions.

The resulting formulation provides a common objective for a range of computational workflows. These include single-sequence scoring, penetration-depth analysis, single-environment optimisation, diverse family generation, dual-environment comparison, and cross-environment compromise optimisation. The same physical framework is therefore used consistently from sequence evaluation through environment-aware sequence design.

A principal advantage of the platform is that environmental conditions are incorporated directly into the scoring formulation rather than being treated solely as annotations associated with the sequence. This enables the same candidate sequence to be evaluated under different physicochemical conditions and allows optimisation objectives to be formulated explicitly in terms of environmental compatibility, discrimination, or compromise.

The web-based implementation combines these computational workflows with random-background benchmarking, helical-wheel visualisation, interactive environmental analysis, and idealised α -helical PDB generation. Consequently, the platform provides an integrated interface through which researchers can move from sequence hypothesis generation to the construction of standardised structural starting models.

Importantly, the resulting scores should be interpreted as predictions of compatibility within the defined Hamiltonian and environmental model, not as direct measurements of biological activity or thermodynamic stability. Experimental validation and, where appropriate, higher-resolution structural or molecular simulation remain necessary to establish the behaviour of designed sequences in real biological systems.

Taken together, *Helix Design Studio* provides a practical environment-conditioned implementation of a mechanistic helix-design framework, making the underlying physical model accessible to systematic sequence exploration while retaining explicit control over the individual physical contributions and environmental parameters.

Data and Code Availability

The web platform is freely accessible at <https://helix.simbioslab.com/>. The complete source code of the web application, together with comprehensive documentation, is available in the public GitHub repository at <https://github.com/SIMBIOS-USC/helix-design-studio-web>.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT-5.x (OpenAI) in order to improve readability, grammar, and language style. After using this tool/service, the authors thoroughly reviewed and edited the content as needed and take full responsibility for the final content of the publication.

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